

Explaining the distribution of preposition stranding across Germanic

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1. Introduction

1.1. *Preposition stranding*

Preposition stranding: movement of the complement of P out of PP.

- 1) The man that I talked **to**
- 2) The man who(m) I talked **to**
- 3) Who(m) did you talk to?
- 4) John I talked **to**, but not Peter.
- 5) He was laughed **at**.

Few of the world's languages have stranding at all (van Riemsdijk 1978; Abels 2003, 2012; Truswell 2009).

Those that do can be divided according to whether they allow stranding generally (i.e. equivalents of (1)-(5)) or only with A'-movement (i.e. do not permit 'pseudopassives' equivalent to (5)); see Abels (2003: 230–231).

1.2. *Aims of the talk*

Big question is how to restrict stranding appropriately.

- Given movement (Internal Merge), why should *any* language disallow stranding?
- Assuming we generally want to rule it out, how do we account for the languages that have it?

There is no generally accepted Minimalist account that answers either question.

My proposal today:

- Antilocality and a functional head between P and D are crucial ingredients (Abels 2012)
- This head is K (Bittner & Hale 1996; Neeleman & Weerman 1999):
 - At PF: realized as morphological case (prototypically: dative)
 - At LF: semantics of location in abstract space
- These ingredients make the correct predictions for the cross-linguistic and language-internal distribution of stranding, at least for the Germanic languages.

2. Analytical approaches

2.1. Movement or no movement?

Major debate in 1970s and 1980s about stranding in early English:

- Movement-derived (involving a null operator or deletion of a moved element): Chomsky & Lasnik (1977); van Riemsdijk (1978: 286–297); Vat (1978); van Kemenade (1984, 1987)
- Non-movement-derived ('unbounded deletion'): Bresnan (1976); Allen (1977, 1980a, 1980b); Maling (1978); Kiparsky (1995); also Grimshaw (1975) on Middle English, Maling (1976) on Old Icelandic, Hoekstra (1995) on Frisian

In modern terms (see Hoekstra 1995, Kiparsky 1995), we can cast this as operator movement vs. resumption:¹

- Movement: $[_{XP} YP_i [X' X \dots [_{PP} [P' P YP_i]]]]$
- Non-movement: $[_{XP} YP_i [X' X \dots [_{PP} [P' P \emptyset_i]]]]$

Recently the debate has been revived:

- Brandner & Bräuning (2013) explicitly propose a movement-free analysis for Alemannic relative clauses with invariant relativizers, apparently independently.
- Pittner (2024) argues against a movement-based account for German R-stranding, based on some apparent problems with the account in Müller (2000) (which, however, relate to the motivation for movement, not the presence or absence of movement itself).

¹ Assuming that resumption is not itself movement-derived (which is likely to be true at least for island-insensitive resumption: see Salzmann 2017: ch. 3 for extensive discussion).

My stance on this:

- In a theory where movement comes for free, convincing arguments are required to show that a phenomenon that looks like movement does *not* involve it (see e.g. Thoms & Walkden 2019).
- Such arguments have not been presented.
- There are also a few counterarguments (see Appendix 1).
- I therefore continue to assume that preposition stranding involves movement (see also van Kemenade 2020 on Old English).

2.2. Approaches to stranding

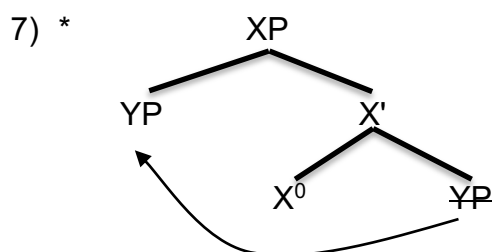
Two main approaches (cf. Truswell 2009); I'll ignore the pseudopassive here:

- Reanalysis theory (Hornstein & Weinberg 1981)
 - Key idea: PP is an island. P⁰ gets 'reanalysed' as forming a constituent with V⁰, as part of the derivation. Complement of PP becomes complement of the verb.
 - Problems:
 - Relies on a questionable contrast between *Who did John speak to Harry about yesterday?* and *Who did John speak to Harry yesterday about?*
 - Conceptually: what is reanalysis? A separate operation in our theory (see Drummond & Kush 2015 for a reanalysis of reanalysis)
 - No evidence for the constituency of P⁰ and V⁰ (and some apparent counter-evidence, e.g. from gapping).
- Escape hatch theory (van Riemsdijk 1978; Abels 2003, 2012)
 - Key idea: PP is an island; what can escape is variable.
 - van Riemsdijk (1978): some languages can use SpecPP as an escape hatch, but others cannot.
 - Abels (2003): the phasehood of PP is parameterized.
 - Problems:
 - Conceptually: parameterization of islandhood adds complexity to our theory.
 - For van Riemsdijk (1978): extraction from PP is *always* possible, as shown by examples like (6) from German (Abels 2003: 211).

- 6) Über welches Thema hast du mich noch mal nach einem Buch gefragt?
 about which topic have you me again after a book asked
 'Which topic did you ask me about a book on again?'

2.3. Abels (2012): an anti-locality account

Anti-locality (Abels 2012: 89; cf. also Abels 2003): movement that is ‘too local’ is not possible.



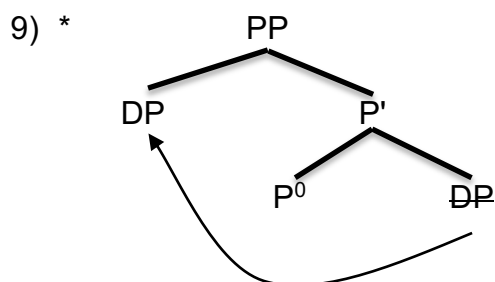
Ingredients of a theory of the ban on stranding (Abels 2012: ch. 7):

- Comp-to-spec antilocality
- PP is a phase

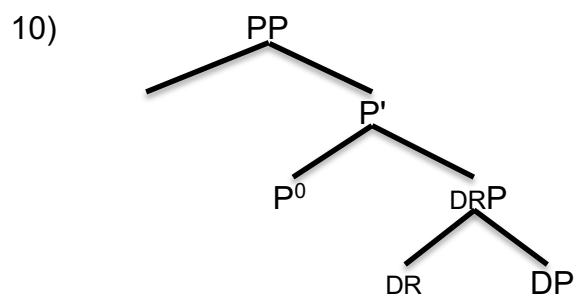
Extraction from PP is available even in languages that disallow stranding (Abels 2012: 206–220).

- 8) De combien (est-ce que) tu as besoin de photos?
 of how.many is-it that you have need of photos
 ‘How many photos do you need?’ (Abels 2012: 212, (55)a,ii)

If PP is a phase, such extraction must proceed via the specifier of PP. But the complement of P is too local to be extracted, hence stranding is ruled out.



How then do we account for the languages in which stranding appears to be possible? Abels (2012): there’s an extra functional projection between PP and DP.

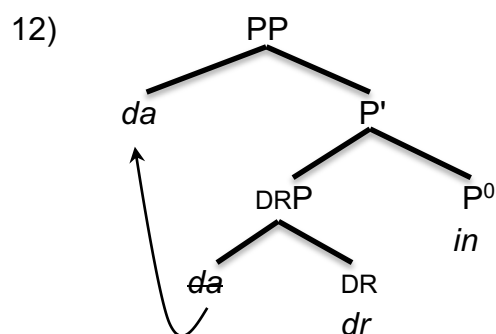


Abels labels this DRP, as his theory is based on R-stranding in German.

11) Da₁ schlafe ich [PP t₁ *(dr)in]
 PRON sleep I DR.in
 'There/It, I sleep in'

Abels's (2012: 231–245) approach to R-stranding in German and Dutch:

- R-words ≠ DP pro-forms, since they interact for locality with locative elements but not pronouns.
- R-words are not complements of P⁰ (cf. Trissler 1993).
- R-words are base-generated as the complement of a lower head which he labels DR (cf. Gallmann 1997).
- The *dr-* in *drin*, *drüber* is the head of the projection below P⁰ which has been incorporated into it. 'DR realizes the case assigned by P' and it 'selects R-words and nothing else' (Abels 2012: 241).



But what about stranding languages like English?

Abels (2012: 245): in these languages, DR is phonologically empty, and it is its DP complement that moves.

Questions that arise:

- a) Is there any independent evidence for this in stranding languages?
- b) What, exactly, is DR? What is its content?
- c) Why is null DR not more widely attested cross-linguistically?

Rest of this talk: leaving aside c), I'll present two case studies that may help to shed light on a) and b), and thus support an account that makes use of a functional projection above DP and below PP.

- Section 3: stranding in Old English (and the other early West Germanic languages)
- Section 4: a KP account and its predictions
- Section 5: impossible strandings in present-day English and German

3. Stranding in Old English

Old English: West Germanic composed before 1150 CE.²

Two main environments for stranding: (some) relative clauses (3.1) and R-words (3.2). What's striking is the form of stranded prepositions (3.3).

Many of the patterns and arguments presented here hold also for Old Saxon and Old High German: these facts are discussed in more detail in Walkden (under revision).

3.1. Relative clauses

Three main types of relative clause in Old English (Traugott 1992: 224–8; Taylor 2014: 467–70):

- a) Formed with a relative pronoun, an inflected form of the demonstrative *se* 'the/that' (case is taken from the relative clause; case attraction possible but rare):

13) *pa twegen fet þa þeo sawul habban sceal*
 the two feet which.MASC.PL.ACC the soul have shall
 'the two feet which the soul shall have' (coadrian, Ad:26:1.61)

- b) Formed with the invariant complementizer *þe* (this is the most common structure):

14) *to his agenum æþele, þe he on geboren wæs*
 to his own country COMP he in born was
 'to his own country which he was born in' (coaelive, +ALS_[Basil]:85.506)

² References to OE examples are given as token IDs from the YCOE (Taylor et al. 2003).

c) Formed with both of the above:

15)Se weig **se** **þe** læt to heofonrice
 the way which.MASC.SG.NOM COMP leads to heaven
 'the way which leads to heaven' (cocathom1,+ACHom_I,_10:264.169.1954)

Stranding is impossible in Old English when a relative pronoun is present in SpecCP, and obligatory when it is not (see Traugott 1992: 230–231; Fischer 1992: 388–389; Taylor 2014: 444–445).

16)ðone eorðlican wisdom **be þam þe** þus awriten is
 the earthly wisdom about which COMP thus written is
 'the earthly wisdom about which is written thus:' (coaelive,+ALS_[Christmas]:227.183)

17)& þone dracan acwealde **þe** we **on** belyfdon
 and the dragon killed COMP we in believed
 'and killed the dragon that we believed in' (cocathom1,+ACHom_I,_37:504.202.7459)

Other, more minor types:

- Zero or contact relative clauses (Harbert 2007: 456: 'virtually unattested')
- Relative clauses formed with invariant complementizer *þæt*
- Infinitival relative clauses with *to*

18)hi næfdon hlaf to etanne
 they NEG-had loaf to eat
 'they did not have bread to eat' (cowsgosp,Mk_[WSCp]:3.20.2353)

These minor types of relative clause (zero, invariant *þæt*, infinitival) also allow stranding, unsurprisingly.

3.2. R-stranding

R-stranding is 'common to all West Germanic languages' according to Harbert (2007: 453), including early West Germanic and Old English in particular. Not specific to relative clauses.³

³ In Old English, something like R-stranding can also occur with personal pronouns:

- ond misenlico wilddeor **him** þær comon **to**
 and various wild-beasts him there came to
 'and various wild beasts there came to him' (comart3,Mart_5_[Kotzor]:Ju2,A.6.887; Alcorn 2009: 434)

There is no particularly good story for this (Wende 1915; Alcorn 2009, 2011), though see Appendix 2.

19) þonne gæþ **þær** swiðe mycel hwil **to**
 then goes there very great while to
 ‘then it will take a great deal of time’ (cocathom2,+ACHom_II,_1:9.214.190)

In Old English, non-R-stranding does not occur in interrogatives (Taylor 2014: 445), with topicalization (Fischer 1992; Taylor 2014: 446, *pace* Abels 2012: 262–3) or in (pseudo)passives.

Generalizations: in Old English, stranding

- may occur with R-movement
- may occur with operator movement
- may *not* occur with an overt extracted complement

Another way of stating this: stranding is permitted only in A'-movement constructions that do NOT involve movement of an overt case-marked full DP.

3.3. Evidence for DR?

Crucial evidence for an Abels-style account of Old English stranding, not adduced in the 1970s–1980s debates, comes from differences in the form of stranded prepositions (Wende 1915; Alcorn 2011).

20) þæt hus þe hiora godas **inne** wæron
 the house COMP their gods in were
 ‘the house that their gods were in’ (coorosiu,Or_6:14.142.1.2975)

Alcorn (2011: 166–191) on Old English (data from the YCOE, Taylor et al. 2003):

- *be* ‘by, concerning’ never strands (N=271), but alternates with *bi*, *bie*, *big*, *bii*, *by*, which does strand (5/13; 38%).
- *for* ‘before, because of’ never strands (N=191; Wende 1915: 14), but alternates with *fore*, which does strand (25/29; 86%).
 - Two examples of *þærfore* ‘therefore’, with final vowel. **þærfor*.
- *betweonum* almost always strands (224/236; 95%), but other forms such as *betweoh* almost never do (20/139; 14%).

Further Old English data:

- *in* ‘in’ is rarely stranded (25/128; 20%), whereas *inne* as P usually is (34/41; 83%), including 25

with *pær* as in (21), 8 with relative clauses as in (20), and one with a personal pronoun (22).⁴

- 21) Gif **pær** wyrms **inne** bið. Hyt **pæt** ut awyrpð.
 if R worms in are it that out throws
 'If there are worms in it, it will drive them out' (coquadru,Med_1.1_[de_Vriend]:7.17.297)
- 22) hweper ænig liflic oroð **him** þa gyt **inne** wære
 whether any living breath him then still in were
 'whether any living breath could still be found in him'
 (cogregdC,GDPref_and_4_[C]:12.276.14.4031)

- Other, rarer forms:
 - *ymb(e)* 'about' is almost always stranded (Alcorn: 121/129; 94%); *ym* is attested twice, neither stranded.
 - *mid* 'with' is rarely stranded (Alcorn: 126/1,252; 10%); *mide* is attested twice, both stranded.
 - *toward/toward/towerd* 'toward' are relatively rarely stranded (10/32, 31%); *towearde* is attested twice, both stranded.

The same facts hold of all types of stranding (relative clauses, R-stranding, etc.).

Table 1: Variation in Old English preposition form (simplified)

Non-stranded form	Stranded form
<i>be</i>	<i>bi, bie, big, bii, by</i>
<i>betweoh, betweox, betwux</i>	<i>betweonum</i>
<i>for</i>	<i>fore</i>
<i>in</i>	<i>inne</i>
<i>mid</i>	<i>mide</i>
<i>toward, toward, towerd</i>	<i>towearde</i>
<i>ym</i>	<i>ymbe</i>

Another relevant fact: *purh* 'through', the only Old English preposition that consistently takes an accusative complement, virtually never strands (Alcorn 2011: 167–169).

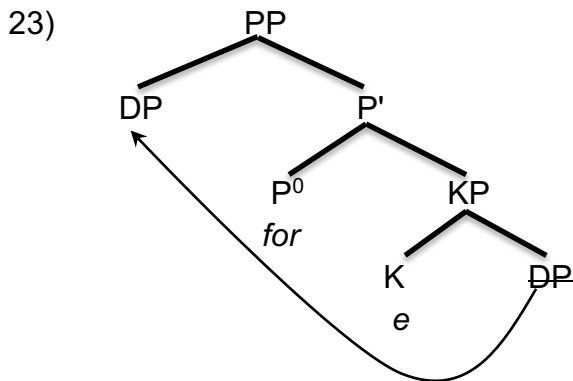
Difficulty: instances of stranding are not always easy to distinguish from verbal particles (Petrova 2021).

Following traditional dictionary treatments, Alcorn (2011) argues that the variation in *be/bi* and *for/fore* in Old English is purely phonological, based on whether the P can cliticize onto a following object. BUT the two variants aren't always phonologically predictable from one another, and this treatment

⁴ There are also some examples in the YCOE that are (mis)tagged as adverbs.

doesn't carry over to *betweenum* (cf. Rossi 2012: 235–243).

Alternative proposal: the stranded forms contain an additional morpheme corresponding to a K head. Direct supporting evidence: in Old English, -e is the dat. sg. ending of strong nouns.⁵ (-um is the dat. pl. ending on all nouns.) K is Abels' DR; recall Abels (2012: 241): 'DR realizes the case assigned by P'. Case-marked 'DPs' are KPs; null operators realize a structure that is no bigger than the complement of K.



A KP analysis for case-marked nominals has long been entertained for Germanic languages, especially for dative case (e.g. Bayer, Bader & Meng 2001). Details of implementation:

- KP will never be able to vacate PP, due to antilocality & the fact that it is the complement of P⁰.
- By contrast, DP can, in principle, move out via Spec,PP.
- Assume further that case-marking on nominal elements other than K (determiners, nouns...) arises via Agree, and that Agree is postsyntactic, taking place after chain reduction.⁶
 - If DP moves without taking the KP layer along with it, chain reduction then bleeds the DP-internal spread of case features.
- Under such a system, a DP that is extracted out of KP cannot be spelled out as case-marked.
- Repair strategies:
 - R-words (Old English *þær*, German *da*), spelling out a caseless DP.
 - Realization of K as dative -e or -um depending on number specification (Old English), or as *dr-* (German), attaching to P⁰.
 - Exceptional PF-deletion of both ends of the chain in Old English relative clauses?

⁵ The presence of these endings must also have an etymological explanation – but this doesn't obviously account for their synchronic distribution. Describing the postpositional use as 'adverbial', as in Pittner (2024), is begging the question, and the richer system of unstressed vowels in Old High German makes it even clearer that we are dealing with case endings; see Walkden (under revision).

⁶ These are nonstandard assumptions, but not wildly so. One option: Agree is an entirely postsyntactic operation (Bobaljik 2008). Another is to decompose Agree as operation into Agree-Link and Agree-Copy, with only Agree-Copy being postsyntactic (Arregi & Nevins 2012: 7–8). In the latter case, chain reduction disrupts the link established by Agree-Link, hence bleeding Agree-Copy of case features onto nominal heads.

Finally, we need a phonological repair process such that derived sequences of vowel plus schwa are resolved into a long vowel sharing the features of the preceding vowel, i.e. /i/ + /ə/ > /i:/.

- This is needed to derive such forms as OE *bi*, *bie*, *big*, *bii*, *by* which plausibly reflect a long vowel.

4. Cross-linguistic implications: morphological case and stranding

This theory predicts, quite generally, that KPs ought to be unstrandable. Insofar as morphological case realizes a K head, there should be a close relation between morphological case and strandability: morphologically case-marked nominals should be unstrandable.

Languages that allow generalized stranding with nominals (Truswell 2009, Abels 2012):

- **English**
- **Norwegian**
- **Swedish**
- Vata
- Gbadi
- Prince Edward Island French

Languages that allow stranding only under A'-movement:

- **Danish**
- **Faroese**
- **Icelandic**
- **Afrikaans** (maybe)
- **Dutch** (???)
- Papiamentu
- São Tomense Creole
- Cape Verdean Creole

These are largely languages without morphological case-marking on normal DPs; Icelandic and Faroese are the only exceptions.

24) Hann spurði **hvern** eg hefði talað **við**.
 he asked who.ACC I had talked to.

'He asked who I had talked to.' (Icelandic; Maling & Zaenen 1985: 199)

Approaches to these facts:

- a) Give up on the connection between case and unstrandability.
- b) Posit that there is something different about morphological case in Icelandic (as opposed to e.g. German).

Since there is independent evidence for b), we'll explore that.

4.1. What's different about case in German and Icelandic?

Three main approaches to the analysis of morphological case (McFadden 2024):

- i. Standard Case Theory: morphological case reflects syntactic Case. (the Vergnaud-Chomsky tradition, now largely abandoned for this purpose)
- ii. KP: Morphological case reflects a rich hierarchy of functional projections in syntax. (locus classicus: Caha 2009, building on the KP tradition)
- iii. Case is morphology (CisM): Morphological case is a PF phenomenon, determined postsyntactically but based on syntactic configuration ('M-case'; e.g. Marantz 1991)

McFadden (2024): we need both approaches ii., KP, and iii, CisM. He argues for a synthesis:

- Inherent case is best captured with ii, KP.
- Structural case is best captured with iii, CisM.

I will also assume that we need both ii. (more underlying structure than just DP) and iii. (realizational rules at PF).

- KP results from selection for differently-sized amounts of structure, and happens syntax-internally. This may be lexically specific. For both Icelandic and German:
 - Indirect objects are first Merged as KPs in Spec,_VAPPLP.
 - P selects KP.
 - Experiencers are universally merged as KPs.
 - Other arguments are merged as DPs.
- Morphological case results from the same general rules in both languages:
 - Structural accusative is assigned to a caseless DP if c-commanded by a local DP in Spec,_vP with no intervener (cf. McFadden 2004: ch. 6)
 - Structural nominative is assigned to a caseless DP elsewhere (cf. Schütze 2001)
 - Case assignment precedes chain reduction/copy deletion.
- The difference between German and Icelandic results from the following rules:
 - (In German) Morphological dative case is assigned to a KP.
 - (In Icelandic) Morphological dative case is assigned to the complement of K.

This likely relates to our first independent difference between German and Icelandic: German, unlike Icelandic, has D-to-K movement (see e.g. Law 1998).

- Evidence: existence of P+D conflated forms such as *im* ‘in.the’, *zur* ‘to.the’, etc. in German.
 - D-to-P movement is needed to generate these.
 - By the Head Movement Constraint, this movement must go via K.
 - The simplest assumption is then that D-to-K movement is obligatory in German whenever K is present, and K-to-P movement likewise whenever P is present.
- In Icelandic, meanwhile, P+D conflated forms are entirely absent (Sirri Björnsdóttir, p.c.), indicating lack of D-to-P movement and thus (probably) of D-to-K movement.

Following Harizanov & Gribanova (2019), I assume that the D-to-P-to-K “movement” in German is a post-syntactic process of morphological Amalgamation.

- German dative determiner forms are necessarily realizations of D+K.
- Icelandic dative determiner forms are realizations of D.

Let’s see how this derives the differences in stranding. Input to the morphological component in both languages (after movement out of PP):

- $[X_P [DP\ D\ NP]_i [X' X \dots [PP [DP\ D\ NP]_i [P' P [K_P [K' K [DP\ D\ NP]_i]]]]]]$

German:

- The higher copies of DP cannot be spelt out as dative, as K is not locally present.
- A repair rule may kick in at VI insertion: the headless DP can be spelled out as *da* (with variation).
- P+K can be spelled out as *dr+P*, as in Abels (2012).

Icelandic:

- The highest copy of DP can be spelt out as dative.
- K remains silent.

This theory predicts our second independently-attested difference between German and Icelandic: dative-marked arguments should be able to occur in positions that are restricted to DPs, such as the subject position Spec,TP. Famously, this is correct: Icelandic has dative subjects (Zaenen, Maling & Thráinsson 1985), German doesn’t, at least not robustly (Barðdal & Eythórsson 2005: 851–859).

4.2. Why does Dutch exist?

Dutch as standardly described behaves exactly like German as far as stranding is concerned. Since Dutch has lost morphological case, this is a potential problem for our account.

But note that this is not the whole story: stranding is more robustly attested in Dutch than generally acknowledged (Klooster 1989; van der Horst & van der Horst 1999; Broekhuis 2013: 295–6).⁷

- a) Bananen houd ik niet **van**
 bananas like I not of
 ‘I don’t like bananas.’
- b) Dat cadeau ben ik blij **mee**/*met.
 that present am I happy with/with
 ‘I’m happy with that present.’

Examples like a) and b) are characteristic of spoken language, and increase in frequency in the second half of the twentieth century (van der Horst & van der Horst 1999: 270), suggesting change in progress.

Broekhuis (2013) suggests an analysis involving left-dislocation and deletion of a resumptive pronoun *daar*. However, this doesn’t predict the lack of intonational break in the examples above.

Celaya Gutiérrez (2020) conducts an acceptability judgement study on potential clitic left dislocations like a) and b) and also *wh*-questions with stranding as in c):

- c) Welke vruchten ben je dol **op**?
 which fruits are you keen on
 ‘Which fruits do you like?’

Strikingly, examples like c) are rated very highly (mean 4.06 on a 7-point scale), *more* so than examples like a)-b) (mean 3.02).

This suggests, I think, the following picture:

- There is no *principled* reason why a language like Dutch shouldn’t have adposition stranding.
- Nevertheless, in more conservative varieties, it doesn’t (beyond R-stranding).
- This is because Dutch still has some sort of D-to-K movement when K is present.
- Acquirers have little evidence for D-to-K.
 - P-K-D forms like *ten* and *ter* ‘to.the’, while archaic, still exist in set phrases.

⁷ I am grateful to Melle Groen for bringing these facts to my attention.

- Indirect negative evidence (absence of stranding) could count as evidence under some assumptions.
- Prescriptive grammarians/schooling disfavour liberal stranding
- Cf. Coopmans & Schippers (2008: 76–77): ‘Dutch children take a while to master the fact that preposition stranding is limited to this specific class of pronouns. Initially, they seem to have a more liberal English-type grammar, with no restriction to particular pronouns whatsoever.’

So perhaps modern standard/written Dutch is a ‘relic’ language type in which the ban on stranding is being semi-artificially maintained.

5. On the semantics of KP: Unstrandable nominals in present-day English

Detour to present-day English, which Abels (2012) claims has null structure (DR) below P and above DP. I’m claiming that this is a KP layer.

Not everything can strand: under certain prepositions and certain types of movement, axial parts (Svenonius 2006), intervals (rather than events), and locations (rather than entities) (Stanton 2016) cannot be stranded.

25)*It was front of the car that the kangaroo was in.

26)*June, we swam in.

27)*10,000 feet is tough to fly to.

Note also the non-strandability of the complement in *at home* and *in prison*:

28)*Home is tough to be at.

29)*Prison is tough to be in.

This might suggest the following:

- The DR head = K = AxPart (Svenonius 2006)?.
- Semantics: identifies set of points in (geographical or temporal) space, based on Ground DP.
- Intervals and locations obligatorily lexicalize more structure, KP.

5.1. Antipronominalization

Non-stranding contexts are also *antipronominalization* contexts in present-day English (Stanton 2016; NB: *not* a general property of intervals and locations), and the same holds for AxParts:

30)*John was in [front of the car]_i, and Mary was in it_i too.

31)*John went swimming in June_i, and Mary went swimming in it_i too.

32)*John flew to [10,000 feet]_i, and Mary flew to it_i too.

The same seems to be true in German for *da*-pronominalization:

33) Zu dem Thema, dazu habe ich eine Meinung.

34)*Zu Weihnachten, dazu esse ich viel.

35) Im Wald, da wandere ich gerne drin.

36)*Im November, da wandere ich gerne drin.

37) Auf der Mainau, darauf gibt es Blumen.

38)*Auf Grund von Bauarbeiten, darauf gibt es Verspätungen.

(see also Helbig 1974: 133).

6. Conclusion

- I have argued that a version of Abels's (2012) antilocality-based account of preposition stranding is the most promising candidate for a general account of the phenomenon, at least as far as Germanic is concerned.
- The mysterious DR head deployed in Abels's analysis is something that we need independently: K.
 - Evidence for a link to morphological case: Old English (and early West Germanic) and, less obviously, German, as shown in section 3.
 - Evidence for a link to location semantics: present-day English unstrandability (section 5).
- Icelandic and Faroese are *prima facie* challenges to this hypothesis, but cease to be so once the nature of their case systems is properly understood (section 4).
- Dutch exhibits more variation than usually acknowledged, but at least some modern spoken Dutch varieties show true stranding (section 4).

Thanks for listening!

References

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Appendix 1: Arguments against the non-movement analysis for Old English

Key empirical prediction: the non-movement analysis predicts freedom from island effects, whereas the movement analysis does not.⁸ Allen (1980b: 264) states that no island violations are found (see also van Kemenade 1987). Kiparsky (1995) claims that island violations occur in Old English with *þe* but not with null resumptives, giving (i) as an example.

- i) Ac for þæm he geneoðde swiþost ofer þone munt **þe**
 but for that he ventured swiftly over the mount comp
 he wiste þæt Flamineus se consulwende that he buton
 he knew that Flamineus the consulthought that he without
 sorgemehte on þæm wintersetle gewunian
 worry might on the winter-quarters dwell
 ‘but because he ventured swiftly over the mountain that he knew that Flamineus the consul
 thought that he might dwell on in winter quarters without care’ (coorosiu, Or_4:8.100.12.2067;
 Kiparsky 1995: 150–151)

But (i) doesn’t contain an island violation (the translation is perfectly grammatical), so still no

⁸ This is generally true, but not for the non-movement account in Bresnan & Grimshaw (1978), who reformulate Subjacency. Locality also can’t be taken to be diagnostic of movement in Minimalist syntax in general, given the availability of Agree, which by hypothesis is local (cf. Adger & Ramchand 2005: 162) – but I’ll set the Agree possibility aside in what follows.

evidence. Given the general rarity of long-distance extraction in Old English in any case, how could we be sure even if we did find a convincing example? (Cf. Chomsky & Lasnik 1977: 498–499.)

Quick search of the YCOE corpus: there are 5 examples of apparent extraction from a PP across at least one finite clause boundary, and no obvious island violations among them.⁹

Abels (2003: 181–192; 2012: 226–231): resumptives are systematically incompatible with comparatives of inequality. His reasoning: resumptive pronouns can only have the denotations that normal pronouns can have (Sharvit 1999), and gaps in comparatives need the reading *x-many books* (see Abels 2012: 227–229).

Frisian, unlike Dutch (and German), allows stranding with comparatives. This means, according to Abels, that Frisian allows true stranding under A'-movement, *pace* Hoekstra (1995), who analyses such examples as involving a null resumptive.

- ii) *Jan heeft meer geld verdiend dan zijn vrouw op gerekend had.
 Jan has more money earned than his wife on counted had
 *intended: 'Jan made more money than his wife had expected.' (Dutch)
- iii) ✓Jan hat mear jild fertsjinne as dat syn frou op rekkene hie.
 Jan has more money earned than that his wife on counted had
 'Jan made more money than his wife had expected.' (Frisian)

OE also allows stranding with comparatives:

- iv) seo is bradre þonne ænig man ofer seon mæge
 she is broader than any man over see may
 'it is broader than any man can see across'
 (OE; Traugott 1992: 225; coorosiu,Or_1:1.16.9.286)
- v) to beteran tidun þonne we nu on sint
 to better times than we now in are
 'in better times than we are in now'
 (OE; Goh 2004: 484; coorosiu,Or_2:5.48.36.938)

These examples are vanishingly rare: only 1 example of stranding found in the YCOE;¹⁰ (v) is tagged as a separable verb *oferseon* rather than as a stranded preposition. BUT if stranding involves a null operator in early Germanic, rather than a resumptive, then this pattern is predicted. If we take this pattern seriously then we have an argument for the movement analysis of Old English invariant

⁹ Though obviously this relies on the relevant examples being correctly annotated in the YCOE (the relative clause in (i) is annotated as an adverbial clause, for instance) and on my query catching all the examples.

¹⁰ There are also 5 examples involving comparatives of equality in the YCOE; Abels argues that comparatives of equality are irrelevant.

relative clauses.

Appendix 2: Old English personal pronoun extraction

What about extracted third person pronouns like (22), repeated here as (39)?

39) hweþer ænig liflic oroð **him** þa gyt **inne** wære
 whether any living breath him then still in were
 ‘whether any living breath could still be found in him’
 (cogregdC,GDPref_and_4_[C]:12.276.14.4031)

Tentative proposal: third person pronouns (at least in Old English) spell out less structure than first- and second-person pronouns.

- They are DP rather than KP pro-forms (or: D⁰ heads licensing a null complement).
- Dative case, prototypically, expresses a location in space (see also section 5).
- This location may be abstract rather than concrete.
 - cf. Landau (2010) on the locative syntax of experiencers.
 - Ackema & Neeleman (2018): person features also operate on an abstract space, with third person the least specified.
- In Old English pronouns, K functions to specify location in person space.
- Compare well-known English facts such as *we the linguists*, *you the linguists* vs. **they the linguists*.